Experiment-8

STUDENT DATAFRAME

	Student_ID	Name	Department	Marks
0	101	Asha	CSE	85
1	102	Rahul	ECE	78
2	103	Priya	CSE	92
3	104	Arjun	ECE	88

HIERARCHICAL INDEXING

		Marks	Attendance
Department	Student_ID		
CSE	101	85	90
	102	90	95
ECE	103	78	85
	104	88	88
CSE	105	92	96
ECE	106	81	89

CSE STUDENTS:

	Marks	Attendance
Student_ID		
101	85	90
102	90	95
105	92	96

STUDENT 102:

Marks	90
Attendance	95

Name: (CSE, 102), dtype: int64

COMBINED DATAFRAME USING CONCAT

	Student_ID	Name	Marks
0	101	Asha	85
1	102	Rahul	78
2	103	Priya	92
3	104	Arjun	88

CONCATENATED ALONG COLUMNS

	Student_ID	Name	Department	Marks	City	Age
0	101	Asha	CSE	85	Hyderabad	20
1	102	Rahul	ECE	78	Vizag	21
2	103	Priya	CSE	92	Chennai	20
3	104	Arjun	ECE	88	Delhi	22

MERGED DATAFRAME

	Student_ID	Name	Department	Marks	Grade
0	101	Asha	CSE	85	A
1	102	Rahul	ECE	78	B
2	103	Priya	CSE	92	A+
3	104	Arjun	ECE	88	A

LEFT MERGE

	Student_ID	Name	Department	Marks	Grade
0	101	Asha	CSE	85	A
1	102	Rahul	ECE	78	B
2	103	Priya	CSE	92	A+
3	104	Arjun	ECE	88	A

RIGHT MERGE

	Student_ID	Name	Department	Marks	Grade
0	101	Asha	CSE	85	A
1	102	Rahul	ECE	78	B
2	103	Priya	CSE	92	A+
3	104	Arjun	ECE	88	A

INNER MERGE

	Student_ID	Name	Department	Marks	Grade
0	101	Asha	CSE	85	A
1	102	Rahul	ECE	78	B
2	103	Priya	CSE	92	A+
3	104	Arjun	ECE	88	A

OUTER MERGE

	Student_ID	Name	Department	Marks	Grade
0	101	Asha	CSE	85	A
1	102	Rahul	ECE	78	B
2	103	Priya	CSE	92	A+
3	104	Arjun	ECE	88	A

MERGING ON INDEX

	Name	Marks	Attendance	Grade
101	Asha	85	90	A
102	Rahul	78	85	B
103	Priya	92	96	A+

ORIGINAL DATAFRAME

	Maths	Science	English
Asha	85	90	88

Rahul 78 82 80
Priya 92 95 91

STACKED DATA
Asha Maths 85
Science 90
English 88
Rahul Maths 78
Science 82
English 80
Priya Maths 92
Science 95
English 91
dtype: int64

UNSTACKED DATA
Maths Science English
Asha 85 90 88
Rahul 78 82 80
Priya 92 95 91

LONG FORMAT
Student Subject Marks
0 Asha Maths 85
1 Asha Science 90
2 Rahul Maths 78
3 Rahul Science 82
4 Priya Maths 92
5 Priya Science 95

PIVOTED DATA
Subject Maths Science
Student
Asha 85 90
Priya 92 95
Rahul 78 82

PIVOT TABLE
Subject Maths Science
Department
CSE 87.5 92.0
ECE 80.0 88.0

WIDE FORMAT

	Student	Maths	Science	English
0	Asha	85	90	88
1	Rahul	78	82	80
2	Priya	92	95	91

LONG FORMAT USING MELT

	Student	Subject	Marks
0	Asha	Maths	85
1	Rahul	Maths	78
2	Priya	Maths	92
3	Asha	Science	90
4	Rahul	Science	82
5	Priya	Science	95
6	Asha	English	88
7	Rahul	English	80
8	Priya	English	91

SORTED HIERARCHICAL INDEX

		Marks	Attendance
Department	Student_ID		
CSE	101	85	90
	102	90	95
	105	92	96
ECE	103	78	85
	104	88	88
	106	81	89

AVERAGE MARKS BY DEPARTMENT

Department
CSE 88.5
ECE 83.0
Name: Marks, dtype: float64